ABDUL REHMAN

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AI engineer with an architecture background: a B.Arch and three years in architectural practice and business development. I build and evaluate AI systems for building regulations, BIM and geodata, and I document where they fail.

EDUCATION

M.Sc. Sustainable Mobilities | Hochschule für Wirtschaft und Umwelt Nürtingen-Geislingen (HfWU) 2024 – 2026

- Thesis, grade 1.1 (very good, German scale): "AI-Assisted Regulatory Intelligence for Built Environment Professionals: Restoring Information Mobility in Stuttgart's Multi-Level Building Governance"
- Supervisors: Prof. Dr. Burkhard Hoppenstedt and Prof. Dr.-Ing. Kirsten David

B.Arch. (Bachelor of Architecture) | NED University of Engineering and Technology, Karachi 2016 – 2021

- Final thesis awarded first position at the university

SELECTED PROJECTS

AI for digital building permitting in Stuttgart | M.Sc. thesis system, deployed as a web application 2025 – 2026

- One question answered from three independent sources, with disagreements shown instead of hidden: retrieval over 12,701 passages of federal, state and municipal regulation text (CrewAI agents, FastAPI, Docker); a YOLOv8 and EasyOCR pipeline that reads zoning plans; ALKIS cadastre lookup with GeoPandas and QGIS
- Evaluated three ways: a benchmark of 25 queries with answers verified by hand, a usability study with 55 participants, and consultation with Stuttgart's building law office and city planning office

Geodata and AI integration: the Weissenhof case study | Independent study 2026

- Cross-checked five official datasets for one site (ALKIS, LoD2, DGM1, DOP20, beta OGC API) with an AI agent working in Blender through MCP. 69 of 69 footprints matched; found a roof that the 3D model gets wrong

Claude and Revit through MCP | Case study 2026

- Tested plain-language queries on a live Revit model with two existing connectors and documented what worked, what failed and what not to trust

vCOO: virtual compliance officer | Research assistant project, HfWU 2025 – 2026

- Built five MCP servers (EU AI Act, GDPR, ISO 27001, COBIT, NIST) with FAISS retrieval and a scripted prototype on a mobility use case

RESEARCH AND TALKS

DBP26, Munich, November 2026: accepted poster, "Restoring Information Mobility in Multi-Level Building Governance"

OGC iDays 2026, Munich, November 2026: accepted presentation

In preparation: journal paper on the evaluation of the Stuttgart system; whitepaper on AI use cases for the construction and real estate sector (co-author, with suisse.ing)

EXPERIENCE

Research assistant (HiWi) | HfWU, Python and AI course for real estate students Jan – Jun 2026

- Developed a four-module set of teaching notebooks (Python fundamentals, sentiment analysis, sales forecasting, computer vision) and assisted the students in class

Research assistant (HiWi) | HfWU, compliance research project Dec 2025 – Mar 2026

- Designed and built the knowledge layer and prototype of the vCOO compliance system

Business Development and Marketing Manager | Disruptive Designs LLC, USA Mar 2023 – Sep 2024

- Built international client acquisition for a US architecture firm: CRM, lead generation, digital tools and automation

Architect | SHA Architects Sep 2021 – Feb 2023

- Design work in coordination with clients, contractors, surveyors and permitting authorities; technical documentation in 2D and 3D

SKILLS AND LANGUAGES

AI and agents: Python · LLMs and vision-language models · retrieval (RAG) · MCP · CrewAI · LangChain · evaluation and benchmark design

Built environment: German planning and building law (BauGB, BauNVO, LBO BW) · zoning plan analysis · ALKIS cadastre data · QGIS · BIM workflows, Revit, IFC

Languages: English (fluent) · German (B1/B2, improving) · Urdu (native)

Certificates: The Complete Agentic AI Engineering Course (Udemy) · Supervised Machine Learning (Coursera) · Advanced Learning Algorithms (Coursera)